



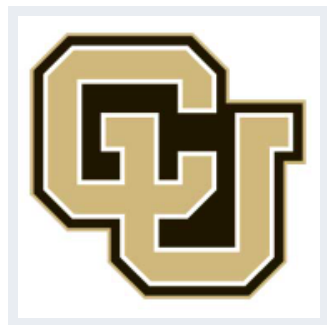
4 Courses

**Cryptography and
Information Theory**

Symmetric Cryptography

**Asymmetric Cryptography
and Key Management**

**Cryptographic Hash and
Integrity Protection**



Oct 7, 2025

Saptarshi Mandal

has successfully completed the online, non-credit Specialization

Applied Cryptography

This specialization is intended for the learners interested in or already pursuing a career in computer security or other cybersecurity-related fields. Through four courses, the learners will cover the security of information systems, information entropy, classical cryptographic algorithms, symmetric cryptography, asymmetric/public-key cryptography, hash functions, message authentication codes, digital signatures, key management and distribution, and other fundamental cryptographic primitives and protocols.

S-Y S
Sang-Yoon Chang

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

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